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Sufficiency-Driven Business Model Canvas: a framework for operationalising sufficiency in a business context

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Abstract: This study applies the Design Research Methodology to explore the concept of sufficiency and identify key characteristics of businesses operationalising sufficiency. The literature review revealed the absence of a universal definition of sufficiency and a noticeable gap between the theoretical understanding and practical application. To address this, the study presents a set of sufficiency-driven business strategies and develops the Sufficiency-Driven Business Model Canvas - a framework that clarifies core elements as well as supports companies in developing sufficiency-driven business models.

Keywords: Sufficiency, Business Model, Circular Economy, Sufficiency-driven business

1. Introduction

The urgency of addressing the global climate crisis and escalating environmental challenges has never been greater. Human activities have accelerated temperature rise, biodiversity loss, and severe ecosystem disruptions, pushing natural systems beyond their capacity to recover (Calvin et al., 2023; Richardson et al., 2023). Research indicates that seven out of nine planetary boundaries have already been crossed, bringing the Earth's system dangerously close to irreversible tipping points (Sakschewski et al., 2025). At the core of these environmental issues lies humanity's reliance on unsustainable, linear economic systems characterised by a 'take-make-dispose' logic that heavily depends on fossil fuels and finite resources (EEA, 2019). To mitigate the most severe environmental impacts, fundamental changes are required in both production and consumption patterns (Calvin et al., 2023). In recent years, the circular economy has emerged as a promising alternative to this linear model (Kirchherr et al., 2023). The circular economy framework seeks to minimise resource use and waste generation by keeping materials and products circulating within the economy for as long as possible (Bocken et al., 2016; Ellen MacArthur Foundation, 2019; European Parliament, 2023). Nonetheless, even with its potential to reduce environmental and social pressures, the benefits of circular approaches are increasingly offset by the continuous rise in production and consumption of new products (Bocken et al., 2022; Freudenreich & Schaltegger, 2020; Metic & Pigosso, 2022). As a response, the concept of sufficiency has gained prominence in sustainability discussions. Sufficiency is positioned as a countermeasure to overconsumption, traditionally understood as a matter of individual consumer responsibility (Beulque et al., 2023; Freudenreich & Schaltegger, 2020). However, focusing solely on consumer behaviour overlooks the significant role that businesses can play. Companies have the potential to act as key enablers of sustainable consumption by implementing sufficiency-oriented business models and strategies (Freudenreich & Schaltegger, 2020). Nevertheless, the role companies play in driving sufficiency through new business operations is still not clear (Bocken et al., 2022). To address this gap, this research seeks to explore how businesses can actively contribute to fostering sufficiency-driven patterns of production and consumption and provide a framework for implementing sufficiency principles into business operations.

2. Research Methodology

This study employs the Design Research Methodology (DRM), a structured framework to integrate theoretical insights with practical application (Blessing & Chakrabarti, 2009). The methodology is organised in four iterative and interconnected stages (Table 1):

Table 1: The key activities performed within the Design Research Methodology.

DRM stage	Key activities
Research Clarification	Exploratory literature review: the concept of 'sufficiency'.
Descriptive Study I	Systematic literature review: sufficiency-driven strategies and business models.
Prescriptive Study (I)	Conceptualisation of a practical support for a sufficiency-driven business.
Descriptive Study II	Expert and user evaluation of the developed support.
Prescriptive Study (II)	Second conceptualisation of the Sufficiency-Driven Business Model Canvas.

Research Clarification (RC) establishes a clear conceptual foundation for sufficiency by consolidating and synthesising the scientific literature. It defines the concept, delineates its theoretical boundaries, and clarifies its relevance within contemporary sustainability discourse and business research.

Descriptive Study I (DS-I) examines the relationship between sufficiency and business practice through a systematic literature review (SLR). The SLR was conducted in the database Scopus in 2024. To address the research question "*How are sufficiency-based business models defined in the literature, and what frameworks and strategies exists?"*, the search string combines three blocks of keywords: (i) sufficiency, (ii) business model, and (iii) sustainability and circularity. The search retrieves 172 documents, including scientific articles, conference papers, book chapters, and books. 12 studies are selected based on two criteria. (1) The research must address sufficiency according to this study's definition, and (2) the research must either present core characteristics of a sufficiency-driven business or present implementable sufficiency-driven strategies for businesses. The identified studies examine cases within a broad range of industry sectors, including electronics, apparel, construction, and household appliances. Characteristics for a sufficiency-driven business is extracted from the literature and clustered in themes within the Business Model Canvas (Osterwalder et al., 2010). The sufficiency-driven business strategies are added to an existing framework presented in the included literature. Furthermore, the SLR highlights theoretical and practical gaps that justify the development of a dedicated operational framework.

Prescriptive Study (PS) translates conceptual advancements into actionable design propositions. Through iterative ideation and refinement, the 'Sufficiency-Driven Business Model Canvas' is developed as a structured tool to operationalise sufficiency within business model design. *Descriptive Study II (DS-II)* evaluates and refines the proposed framework. The evaluation consists of (1) expert assessment involving five specialists in business development and sustainable development, and (2) application in three potential user scenarios within the apparel industry: an educational setting, an established company, and a start-up. Feedback regarding applicability, usability, and effectiveness informs further refinement through an additional PS iteration. By following the DRM iteratively, this study ensures that the developed concepts are both scientifically grounded and applicable in practical business contexts.

3. The Concept of 'Sufficiency'

The literature demonstrates that the notion of sufficiency is interpreted across a wide range of academic disciplines (e.g., ecology, economics, and philosophy) (Jungell-Michelsson & Heikkurinen, 2022) and still lacks a universally accepted definition. Despite this conceptual diversity, several recurring themes can be identified. At its core, sufficiency focuses on upstream strategies for waste prevention and is situated at the upper levels of the waste hierarchy, emphasising the principles of *Rethink, Reduce,* and *Refuse* as key levers for limiting resource use (Bocken et al., 2022; Sarokin & Bocken, 2024). Additionally, sufficiency has emerged as a critical response to unsustainable patterns of production and consumption and is increasingly recognised as a complementary strategy to *efficiency* and *consistency* within the circular economy framework (Beulque et al., 2023; Bocken et al., 2016; Gossen & Kropfeld, 2022; Stahel, 2001). Freudenreich and Schaltegger (2020) differentiate these strategies by describing efficiency as '*making and doing more with less*,' whereas sufficiency aims to '*make do with less*.' This distinction underscores sufficiency's transformative potential, as it goes beyond optimisation to question the fundamental necessity of resource use. Although resource reduction is inherently part of the circular economy, sufficiency reinforces its foundational intent by advocating for an absolute reduction in material and energy throughput through lower consumption and production levels (Gossen & Kropfeld, 2022; Sarokin & Bocken, 2024; Soulis & Mont, 2024). Sufficiency, therefore, embodies the idea of '*enoughness*' - ensuring a state of resource consumption that is within the limits of planetary boundaries (Beyeler & Jaeger-Erben, 2022; Bocken et al., 2022; Garcia-Ortega et al., 2023; Niessen & Bocken, 2021). Beyond its material dimension, sufficiency is closely linked to political ecology, as it addresses issues of social justice and equitable distribution in the face of accelerating globalisation and industrialisation (Jungell-Michelsson & Heikkurinen, 2022). It also seeks to slow down *social metabolism* - the intense flow of energy and materials that drives ecological degradation (Jungell-Michelsson & Heikkurinen, 2022) - by curbing overconsumption (Beyeler & Jaeger-Erben, 2022; Bocken et al., 2022; Bocken & Short, 2016; Gossen & Kropfeld, 2022; Kelleci & Yıldız, 2021). In doing so, sufficiency aligns with the field of ecological economics and resonates with degrowth principles, challenging dominant growth-oriented economic paradigms (Jungell-Michelsson & Heikkurinen, 2022). Furthermore, it connects to ecological philosophy, particularly *sufficientarianism*, which emphasises fulfilling the basic conditions required for a good life for all (Jungell-Michelsson & Heikkurinen, 2022).

In summary, to capture the intertwined social and ecological dimensions of the sufficiency concept, this paper proposes the following definition: "*Sufficiency aims to limit resource use by reducing overconsumption to levels compatible with social well-being within the planetary capacity.*"

4. Results

This section presents the main findings of the study, including the application of sufficiency within a business context. It outlines the results of the SLR on sufficiency-driven business and the strategies identified in the Descriptive Study I, as

well as the resulting framework for operationalising sufficiency within organisations developed through the Prescriptive Study and Descriptive Study II.

4.1. A Sufficiency-Driven Business

Building on the conceptual framework of sufficiency, it is essential to explore how these principles can be operationalised in a business context. Translating sufficiency into business practice gives rise to the notion of a *sufficiency-driven business model*, which redefines how businesses create and deliver value while remaining within the ecological limits (Beyeler & Jaeger-Erben, 2022; Bocken et al., 2022). The central aim of a sufficiency-driven business model is to reduce resource use by moderating or lowering consumption levels while continuing to deliver genuine value to customers through the employment of sufficiency-driven business strategies (Beyeler & Jaeger-Erben, 2022; Bocken et al., 2022). Sufficiency-oriented businesses can be understood as operating on two interrelated levels (Bocken & Short, 2016; Gossen & Kropfeld, 2022). First, such businesses seek to meet authentic human needs rather than stimulate artificial or excessive wants, which involves creating enabling conditions that support consumers in maintaining a sufficient level of consumption (Beyeler & Jaeger-Erben, 2022; Bocken & Short, 2016). Businesses must actively shape both their market environment and their value propositions. This requires designing products and systems that slow resource cycles by prioritising longevity, durability, and ease of repair, and by providing complementary services that support continued use. In essence, sufficiency-driven businesses aim to create long-lasting products and maintain strong service relationships that help customers retain and care for these products over time, positioning this type of business in the higher levels of the waste hierarchy, placing the designer central in the transition to a sufficiency-driven business. Second, consumers retain agency in this process, as the decision to purchase - or to refrain from purchasing - ultimately lies with them. Consumers are thereby exerting direct influence on slowing down the material throughput and resource demand. This dual structure sets sufficiency-driven business models apart from conventional ones, which typically prioritise profit maximisation above all else (Bocken et al., 2022). In the literature, sufficiency-driven businesses are described both in terms of the elements and characteristics that constitute such organisations and the employable strategies that promote sufficiency-driven consumption and production (Bocken et al., 2022; Niessen & Bocken, 2021).

4.1.1. Characteristics and Elements of a Sufficiency-Driven Business

A defining feature of sufficiency-oriented businesses is the alignment of their overarching purpose with both social and environmental objectives (Bocken et al., 2022). This alignment requires a clear distinction between products and services that genuinely fulfil essential human needs and those that merely stimulate new desires (Beyeler & Jaeger-Erben, 2022). Achieving such coherence across the organisation depends on having a **governance or ownership** structure that allows a high degree of autonomy and limited obligations to increase shareholder value (Beyeler & Jaeger-Erben, 2022; Bocken et al., 2022; Soulis & Mont, 2024). Increasing the possibility for integrating the sufficiency principles throughout the value chain through an appropriate resource strategy (Gossen & Kropfeld, 2022; Tunn et al., 2019). Bocken et al. (2022) found that traditional governance mechanisms can function effectively. However, Beyeler and Jaeger-Erben (2022) argue that embedding organisational purpose and values directly into decision-making and control processes enhances alignment with sufficiency principles. Moreover, **participation in networks** that support sufficiency-driven operations is vital, as collaboration fosters learning, legitimacy, and resilience (Bocken et al., 2022). Financially, sufficiency-oriented businesses tend to embrace a **slow-growth philosophy** focused on long-term impact rather than short-term profit (Beyeler & Jaeger-Erben, 2022; Bocken et al., 2022). This principle extends to scaling strategies, where growth should never compromise the organisation's fundamental mission (Beyeler & Jaeger-Erben, 2022; Bocken et al., 2022). Instead of increasing the volume of units sold, expansion should focus on reaching new market segments or deepening societal impact (Freudenreich & Schaltegger, 2020) e.g., through revenue models that move towards product-service-systems (Tunn et al., 2019). To uphold these principles, transparent measurement and documentation of the organisation's **social and environmental performance** are essential (Bocken et al., 2022) to keep the objective of a reduced consumption level centred (Tunn et al., 2019). Operating a sufficiency-driven business also requires a **holistic and inclusive approach** that engages all actors within the value network (Beyeler & Jaeger-Erben, 2022; Kropfeld & Reichel, 2021). Critical organisational competencies include cultivating **transparency of both operations and impact**, increasing the awareness of environmental and social impacts and establishing a culture of sharing of **sufficiency-related knowledge and competencies** throughout the value chain – both up- and downstream (Beyeler & Jaeger-Erben, 2022; Freudenreich & Schaltegger, 2020; Gossen & Kropfeld, 2022; Kropfeld & Reichel, 2021). Due to the fairly new business approach, it is essential to implement continuous **learning and feedback mechanisms** internally in the organisation to adapt to new modes of operation (Beyeler & Jaeger-Erben, 2022) as well as support **user involvement** in the business development (Beyeler & Jaeger-Erben, 2022; Kropfeld & Reichel, 2021). The active involvement of diverse upstream- and downstream stakeholders further strengthens and refines business practices, ensuring they remain socially and environmentally grounded (Beyeler & Jaeger-Erben, 2022; Kropfeld & Reichel, 2021; Soulis & Mont, 2024). Further, sufficiency-driven businesses must work to establish social norms that prioritise **ecological integrity and collective well-being** (Beyeler & Jaeger-Erben, 2022; Kropfeld & Reichel, 2021). These values are central for a sufficiency-driven business and should be embedded throughout the organisation's operations and relationships. Importantly, adopting a sufficiency orientation requires patience - progress may be

gradual, but sustained commitment and a long-term perspective are essential for nurturing systemic change towards a sufficiency-based circular economy (Beyeler & Jaeger-Erben, 2022; Kropfeld & Reichel, 2021). Finally, a sufficiency-driven approach is characterised through its engagement of downstream actors through the promotion of new social norms of e.g., creating **emotional attachment** to materials or products or creating a sense of **community** among customers that further strengthens feelings of trust and loyalty to the business or emphasising positive meanings of reduced consumption (Gossen & Kropfeld, 2022) as well as re-thinking the customer relations in terms of needed **customer effort** and considerations on the customer role – active or passive (Tunn et al., 2019).

The literature identifies essential building blocks for sufficiency-driven businesses but lacks a holistic framework for what constitutes a sufficiency-driven business model with focus on the applicability of this defined, but still unexplored, way of operationalising a business. The identified characteristics are therefore gathered in a framework building on the well-established Business Model Canvas (BMC) by Osterwalder et al. (2010), a widely recognised tool for business model innovation and development (Osterwalder et al., 2010). The adaptation process involved embedding the key characteristics identified as essential for sufficiency-driven business operations, which include additions to existing blocks of the BMC as well as the introduction of completely new blocks - 'Purpose', 'Organisation', 'Scale-up', and 'Ecological and Social impacts'. The framework is presented in Figure 1.

Figure 1: The elements and characteristics of a sufficiency-driven business.



4.1.2. Sufficiency-Driven Business Strategies

A range of frameworks in the literature seek to conceptualise and categorise strategies for sufficiency-driven business models. To consolidate the strategies identified across these different approaches, the Business for Sufficiency framework developed by Niessen and Bocken (2021) serves as the foundation, as it effectively synthesises key insights from earlier models. This study has built upon this framework by adding additional sufficiency-driven strategies identified in the literature review from eight studies (Table 2). The strategies presented as sufficiency-driven strategies are not fundamentally new but can rather be seen as iterative principles of established circular strategies. The over-representation of Rethink strategies and low representation of Refuse strategies highlights the role companies can play in a sufficiency-driven circular economy: companies can enable and indirectly influence consumer choices and consumption but are not able to directly compel them to refuse or reduce consumption.

Table 2: Sufficiency-driven business strategies framework based on the Business for Sufficiency framework (Niessen & Bocken, 2021).

RETHINK / Consume differently	REDUCE – Consume less	REFUSE – Don't (over)consume
Less clutter / simplified & less		
No ownership (N.O) e.g. rental, sharing, leasing *(Bocken & Short, 2016; Garcia-Ortega et al., 2023; Niessen & Bocken, 2021).*	Demand reduction service e.g. advice for lower energy use *(Bocken & Short, 2016; Niessen & Bocken, 2021; Soulis & Mont, 2024).*	No ownership *(Beulque et al., 2023; Bocken & Short, 2016).*
Personalised production e.g. tailored clothing *(Niessen & Bocken, 2021; Soulis & Mont, 2024).*	N.O + price incentive e.g. pay-per-wash+ higher degree price *(Bocken & Short, 2016; Garcia-Ortega et al., 2023; Niessen & Bocken, 2021).*	Question consumption e.g. Anti-Black Friday campaigns *(Gossen & Kropfeld, 2022; Niessen et al., 2023; Niessen & Bocken, 2021; Soulis & Mont, 2024).*
Premium price *(Bocken & Short, 2016; Gossen & Kropfeld, 2022; Soulis & Mont, 2024).*	Premium price *(Bocken & Short, 2016; Gossen & Kropfeld, 2022; Soulis & Mont, 2024).*	Moderating sales e.g. no commission, no sales, no aggressive marketing *(Beyeler & Jaeger-Erben, 2022; Bocken & Short, 2016; Garcia-Ortega et al., 2023; Gossen & Kropfeld, 2022; Niessen & Bocken, 2021; Soulis & Mont, 2024).*
Green alternative e.g. packaging-free *(Niessen & Bocken, 2021)*		
Less speed / slower & more reliable		
Provide information about product repairability and durability *(Beulque et al., 2023)*	Provide information about product repairability and durability *(Beulque et al., 2023)*	Provide information about product repairability and durability *(Beulque et al., 2023)*
Product purchasing based on repairability and durability *(Beulque et al., 2023)*	Product purchasing based on repairability and durability *(Beulque et al., 2023)*	Product purchasing based on repairability and durability *(Beulque et al., 2023)*
Repairable and durable products value proposition *(Beulque et al., 2023; Beyeler & Jaeger-Erben, 2022; Bocken & Short, 2016; Freudenreich & Schaltegger, 2020; Garcia-Ortega et al., 2023; Soulis & Mont, 2024)*	Repairable and durable products value proposition *(Beulque et al., 2023; Beyeler & Jaeger-Erben, 2022; Bocken & Short, 2016; Freudenreich & Schaltegger, 2020; Garcia-Ortega et al., 2023; Soulis & Mont, 2024)*	Repairable and durable products value proposition *(Beulque et al., 2023; Beyeler & Jaeger-Erben, 2022; Bocken & Short, 2016; Freudenreich & Schaltegger, 2020; Garcia-Ortega et al., 2023; Soulis & Mont, 2024)*
Facilitate reuse *(Freudenreich & Schaltegger, 2020; Garcia-Ortega et al., 2023)*	Consultancy services / Continuous wear and use of products *(Garcia-Ortega et al., 2023)*	Consultancy services / Continuous wear and use of products *(Garcia-Ortega et al., 2023)*
Green alternative e.g. durable, modular, long-life *(Beyeler & Jaeger-Erben, 2022; Bocken & Short, 2016; Niessen et al., 2023; Niessen & Bocken, 2021)*	Promoting product value by emphasising convenience and positive aspects of extending product use *(Garcia-Ortega, 2023)*	Promoting product value by emphasising convenience and positive aspects of extending product use *(Garcia-Ortega et al., 2023)*

<table>
<tr><td>Emotionally upgradable / Upgradable to changing taste (Garcia-Ortega et al., 2023)</td><td>Emotionally upgradable / Upgradable to changing taste (Garcia-Ortega et al., 2023)</td><td>Question consumption / question needs, e.g. to fly (Gossen & Kropfeld, 2022; Niessen & Bocken, 2021; Soulis & Mont, 2024)</td></tr>
<tr><td>Assessment of products' repairability and durability (Beulque et al., 2023)</td><td>Assessment of products' repairability and durability (Beulque et al., 2023)</td><td>Narrative interventions, storytelling (Gossen & Kropfeld, 2022)</td></tr>
<tr><td>Repairability and durability eco-design requirements and support services suppliers (Beulque et al., 2023)</td><td>Repairability and durability eco-design requirements and support services suppliers (Beulque et al., 2023)</td><td rowspan="11">Avoid fostering the pleasure of shopping just for fun or to show status (Garcia-Ortega et al., 2023; Gossen & Kropfeld, 2022; Soulis & Mont, 2024)</td></tr>
<tr><td>Selection of suppliers based on repairability and durability (Beulque et al., 2023)</td><td>Selection of suppliers based on repairability and durability (Beulque et al., 2023)</td></tr>
<tr><td>Co-creation / Involve users in design (Beyeler & Jaeger-Erben, 2022; Garcia-Ortega et al., 2023)</td><td>Co-creation / Involve users in design (Beyeler & Jaeger-Erben, 2022; Garcia-Ortega et al., 2023)</td></tr>
<tr><td>Provide information on the product's impact (Garcia-Ortega et al., 2023)</td><td>Provide information on the product's impact (Garcia-Ortega et al., 2023)</td></tr>
<tr><td>On-demand production (Soulis, Mont, 2024)</td><td>On-demand production (Soulis, Mont, 2024)</td></tr>
<tr><td>Pop-up stores that facilitate that users can try on prototypes of clothing for pre-order (Beyeler & Jaeger-Erben, 2022; Soulis & Mont, 2024)</td><td>Pop-up stores that facilitate that users can try on prototypes of clothing for pre-order (Beyeler & Jaeger-Erben, 2022; Soulis & Mont, 2024)</td></tr>
<tr><td>Design products, services for everyone’s needs (social justice, inclusion) (Beyeler & Jaeger-Erben, 2022; Soulis & Mont, 2024)</td><td>Life extension service e.g. repair, refurbish, etc. service (Beyeler & Jaeger-Erben, 2022; Garcia-Ortega et al., 2023; Niessen & Bocken, 2021)</td></tr>
<tr><td>Personalised production e.g. modularity, timeless design (Niessen et al., 2023; Niessen & Bocken, 2021)</td><td>(Facilitate) Reuse / Easy-to-maintain (Bocken & Short, 2016; Freudenreich & Schaltegger, 2020; Garcia-Ortega et al., 2023)</td></tr>
<tr><td rowspan="3">Reuse e.g. resale, refurbish, repair, facilitate reuse, ease to maintain (Bocken & Short, 2016; Freudenreich & Schaltegger, 2020; Garcia-Ortega et al., 2023; Niessen & Bocken, 2021)</td><td>Long product warranties e.g. lifelong warranty + free repair (Niessen et al., 2023; Niessen & Bocken, 2021)</td></tr>
<tr><td>Lifetime alterations on clothing (Garcia-Ortega et al., 2023; Soulis & Mont, 2024)</td></tr>
<tr><td>Avoid fostering the pleasure of shopping just for fun or to show status (Garcia-Ortega et al., 2023)</td></tr>
<tr><td colspan="3">Less distance / regional & disentangled</td></tr>
<tr><td>Green alternative e.g. local or regional products (Niessen & Bocken, 2021; Soulis & Mont, 2024)</td><td>Short-distance promotion / Encourage local consumption (Niessen & Bocken, 2021; Soulis & Mont, 2024)</td><td rowspan="3">Question consumption / question needs, e.g. to fly (Niessen & Bocken, 2021; Soulis & Mont, 2024)</td></tr>
<tr><td rowspan="2">Open-source creation e.g. open-source that makes it possible for the user to produce, alter or repair themselves (Beulque et al., 2023; Freudenreich & Schaltegger, 2020)</td><td>Life extension service (Beulque et al., 2023; Beyeler & Jaeger-Erben, 2022)</td></tr>
<tr><td>N.O. + price incentive (Beulque et al., 2023)</td></tr>
<tr><td colspan="3">Less market / beyond commerce</td></tr>
</table>

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<tr><td>Open-source creation e.g. 3D-printing spare parts (Freudenreich & Schaltegger, 2020; Niessen & Bocken, 2021)</td><td>Education for repair and reuse e.g. repair guides and training (Freudenreich & Schaltegger, 2020; Niessen & Bocken, 2021; Soulis & Mont, 2024)</td><td rowspan="3">Education for self-sufficiency e.g. training for gardening and DIY (Freudenreich & Schaltegger, 2020; Niessen & Bocken, 2021; Soulis & Mont, 2024)</td></tr>
<tr><td rowspan="2">Exchange platforms e.g. on-/off-fline, resale, swapping (Garcia-Ortega et al., 2023; Niessen & Bocken, 2021)</td><td>Exchange platforms e.g. knowledge share for DIY repair (Freudenreich & Schaltegger, 2020; Niessen & Bocken, 2021)</td></tr>
<tr><td>Life extension service (Niessen and Bocken, 2021)</td></tr>
<tr><td colspan="3">Design e.g. timeless, for repair, modularity, emotional (Freudenreich & Schaltegger, 2020; Garcia-Ortega et al., 2023; Niessen & Bocken, 2021)</td></tr>
<tr><td colspan="3">Awareness-raising e.g. social-media campaign, podcast (Gossen & Kropfeld, 2022; Niessen & Bocken, 2021)</td></tr>
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4.2. The Sufficiency-Driven Business Model Canvas

The case studies presented in the identified literature on sufficiency-driven business models - Eileen Fisher, Decathlon, VAUDE, and Patagonia (Bocken et al., 2022) - have incorporated sufficiency principles, but are simultaneously promoting higher consumption through, e.g., sales incentives and discount campaigns, contradicting the essence of sufficiency. This gap between the theoretical definition and practical application of sufficiency-driven business models indicates that most examples remain conceptual rather than fully realised, revealing that the translation of sufficiency principles into holistic business practices is limited. In the Prescriptive Study, the previously presented framework is therefore adapted into a practical tool that can facilitate the integration and practical application of sufficiency principles within companies.

The resulting tool, the Sufficiency-Driven Business Model Canvas (abbr. SD-BMC), is illustrated in Figure 2. The new elements (highlighted in orange) represent the expanded set of considerations required for businesses transitioning toward sufficiency-driven operations. Collectively, they highlight both the complexity and the degree of organisational commitment necessary to move beyond traditional profit-oriented business logic. The SD-BMC demonstrates that the successful implementation of sufficiency principles demands full organisational alignment around a clear purpose, extending across internal processes and the wider value chain. It demonstrates further that the central transition lies in the re-design of value propositions for consumers as it is this block that embodies the overall aim of sufficiency, namely, to reduce consumption to a sufficient level. Certain conventional business activities may be inherently incompatible with sufficiency strategies, underscoring the need for careful evaluation and redesign of existing practices. Importantly, sufficiency should not be approached as an isolated initiative or a separate business function. Rather, it must be embedded holistically - integrated as a guiding mindset that permeates all levels of the organisation and informs every aspect of business activity. Following expert evaluation and user testing conducted in the Descriptive Study II, this aspect was further strengthened through the introduction of 'sufficiency statements'. These short introductory statements guide users by clarifying the key considerations relevant to particular building blocks of the SD-BMC. Additionally, the evaluation of the tool disclosed the importance of integrating concrete examples to enhance usability as well as the need to highlighting the questions that require additional attention as they are unique to a sufficiency-driven business.

Further, the sufficiency-driven strategies presented in Table 2 complement the SD-BMC by providing additional actionable guidance for implementation. They can serve both as inspiration and as concrete strategic options for realising the vision and objectives defined through the SD-BMC.

5. Discussion of Sufficiency in Practice

Adopting the SD-BMC wishes to extend the implications beyond strategic decisions to the practical work of design, where sufficiency must be actively embedded in business practices. Within a sufficiency-driven business, design shifts from being primarily a tool for generating profit and stimulating consumption to a means of enabling responsible and meaningful use. Designers must rethink the purpose of design and move from products that encourage desire to developing solutions that genuinely meet human needs. This requires integrating sufficiency-driven design principles such as durability, emotional attachment, as well as designing systems that support long-term use and shared value creation. Ultimately, designers become key actors in facilitating sufficient consumption by creating offerings that allow consumers to live well with less.

As designers work to translate sufficiency principles into tangible products and experiences, an equally important challenge lies in translating the values into market communication that resonates with consumers. While the term, *sufficiency*, holds strong conceptual clarity within academic discourse, its current framing - often associated with notions of limitation, restriction and '*enoughness*' - can appear unattractive or even counterintuitive to consumers accustomed to consumption

Figure 2: The Sufficiency-Driven Business Model Canvas (SD-BMC).



as a form of self-expression and identity-building. Contemporary purchasing behaviour is frequently motivated by emotional, social, and aspirational drivers such as status, novelty, and pleasure, which conflict with how sufficiency is commonly presented in scholarly and policy contexts. For sufficiency-oriented companies to succeed commercially, they may therefore need to reframe and rebrand the narrative around sufficiency. Rather than communicating the terms of *less* or *limitation*, businesses can emphasise the positive dimensions of sufficiency - such as *quality over quantity*, *authenticity*, *well-being*, and *meaningful consumption*. Positioning sufficiency as a lifestyle choice that enhances value, connection, and purpose can make it more resonant with contemporary consumers. In doing so, companies not only increase their market acceptance but also contribute to reshaping social norms around what constitutes desirable and responsible consumption.

However, advancing a sufficiency-driven business approach cannot occur in isolation. Transforming the prevailing high-speed consumption paradigm requires systemic change involving multiple actors across the value chain (Bocken et al., 2022; Freudenreich & Schaltegger, 2020; Iten et al., 2024; Ozdamar Ertekin& Atik, 2015). While individual firms can design products, services, and business models that enable and encourage more moderate consumption, their influence remains limited to a certain level. Companies cannot compel consumers to adopt frugal consumption patterns, nor can they compel competitors or partners to align with sufficiency principles. Instead, progress depends on active collaboration, shared learning, and alignment among suppliers, policymakers, NGOs, and consumers. In this sense, companies play a crucial role as facilitators and exemplars - demonstrating that commercial viability can coexist with reduced material throughput and responsible consumption.

5.1 Limitations and Future Work

A main limitation of this work is the relatively few frameworks used as the foundation for the SD-BMC due to the newness of the field of sufficiency. The frameworks build on analysis of cases that are identified to be partly sufficiency-driven as they incorporate strategies that a counterworking the aim of sufficiency. Additionally, this framework should be further validated, in terms of its practical usability, usefulness, and business viability. The research has primarily focused on developing a strong theoretical foundation for the emerging field of sufficiency-driven businesses, and while the proposed framework has been evaluated by academic experts and in an educational setting; thus, testing with practitioners from business contexts remains limited.

Future research should place greater emphasis on elaborating the scaling-up dimension of the SD-BMC, as financial viability remains a crucial factor when transforming existing companies or establishing new sufficiency-driven companies. Further investigation into this area would enhance understanding of how sufficiency-based business models can remain competitive and financially sustainable in the long term. Strengthening the financial dimension of sufficiency strategies requires empirical insights from companies that have successfully implemented these models. Conducting qualitative studies - such as interviews or case analyses - with these businesses could help identify best practices, reveal effective financial mechanisms, and offer practical guidance on how to develop a compelling business case for sufficiency.

Additionally, this study does not address the quantification of what constitutes a sufficient level of consumption within planetary boundaries, nor does it determine the threshold required to ensure social well-being. This limitation constrains the framework's ability to fully operationalise its core ambition - enabling businesses to reduce consumption to levels that ensure social well-being within the planetary capacity.

6. Conclusion

This paper has examined the concept of sufficiency as presented in academic literature and explored how it can be operationalised in a business context.

Building on this conceptual foundation, the study adapts the well-established Business Model Canvas (Osterwalder et al., 2010) into a *Sufficiency-Driven Business Model Canvas*, designed to support companies in translating sufficiency principles into actionable strategies. The research contributes theoretically by advancing conceptual clarity around the notion of sufficiency - positioning it as a complementary yet distinct paradigm within sustainability and circular economy research. It identifies the core dimensions of sufficiency and outlines the defining characteristics of sufficiency-driven business models. From an empirical perspective, the study contributes by translating these insights into a practical framework that enables organisations to assess, design, and implement sufficiency-oriented strategies. The SD-BMC serves as both a diagnostic and developmental tool, facilitating the integration of sufficiency principles across business functions, governance structures, and value-creation processes.

By bridging conceptual understanding with practical application, this work contributes to the growing body of knowledge on how businesses can act as catalysts for sufficiency-driven production and consumption. Ultimately, it offers a pathway for aligning business practices with planetary boundaries and social well-being, thereby supporting the transition towards a more sustainable and equitable economic system.

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